ACADEMIC RESEARCH PROJECT (MBA-WILP)

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE PILANI

EC1 Experiential Learning – **Situated Learning Project**

AI Application Ecosystem (S1-25_MBACCZG580)

PHASE 1

Problem Discovery, Business Context & Opportunity Framing

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Executive Summary

This report addresses a critical operational challenge in the RFQ (Request for Quotation) response process within Tata Technologies' automotive engineering services domain. The current approach relies heavily on manual coordination across multiple teams, leading to delays, miscommunication, and accountability gaps. These inefficiencies impact proposal accuracy, turnaround time, and ultimately, business competitiveness.

The proposed AI-enabled solution aims to streamline this workflow through document intelligence, automated routing, predictive prioritization, and knowledge reuse. By improving efficiency, accuracy, and transparency, the initiative is expected to enhance proposal quality, reduce operational risks, and increase win rates for high-value opportunities.

Context

This assignment is part of the MBA-WILP Experiential Learning Project under the AI Application Ecosystem course at BITS Pilani. The objective is to apply AI concepts to real-world business problems, demonstrating managerial understanding of AI-driven transformation.
The chosen problem—RFQ coordination and response preparation—emerges directly from my role as Engagement Manager at Tata Technologies, where I manage cross-functional teams, validate cost estimates, and ensure timely delivery of proposals. The complexity and strategic importance of RFQs make this an ideal use case for exploring AI's potential to enhance efficiency, accuracy, and decision-making in high-value business processes.

Acknowledgement

I would like to express my gratitude to my faculty mentor and academic advisors at BITS Pilani for their guidance throughout this assignment. This work has been completed independently as part of the MBA-WILP program requirements.

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Abbreviation Table

Abbreviation	Full Form
RFQ	Request for Quotation
OEM	Original Equipment Manufacturer
CFT	Cross-Functional Team
SME	Subject Matter Expert
CM%	Contribution Margin Percentage
TTL	Tata Technologies Limited
AI	Artificial Intelligence
RASIC	Responsible, Accountable, Support, Informed, Consulted
HW-SW	Hardware-Software
JSON	JavaScript Object Notation
CSV	Comma-Separated Values
PDF	Portable Document Format
KPI	Key Performance Indicator

1. Your Organizational Context

Name of the organization: TATA Technologies GmbH

Industry and business domain: Engineering services for the automotive industry.

Job title and department: Engagement Manager, Business Development / Delivery.

Day-to-day responsibilities:

- Respond to RFQs from OEM clients and coordinate bidding processes.
- Manage cross-functional teams (CFT) and SMEs for technical and commercial inputs.
- Ensure project delivery and maintain Contribution Margin (CM%).
- Track proposal progress, resolve bottlenecks, and ensure compliance.

Typical Decision-Making Tasks:

- Prioritize RFQs based on strategic importance.
- Validate cost estimates and timelines.
- Manage accountability and resource allocation.
- Approve final proposal content and pricing strategies.

2. Problem Identification (From Daily Work Only)

Recurring problem:

- Lack of a standardized and efficient process for RFQ coordination and response preparation.

Who is affected:

- Internal teams (Engineering, Finance, Procurement), Engagement Managers, and indirectly OEM clients.

Pain points:

- Delays in RFQ submission and missed deadlines.
- Miscommunication and skipped details during meetings.
- Incorrect interpretation of inputs → flawed estimation.
- Excessive manual effort for large RFQs.

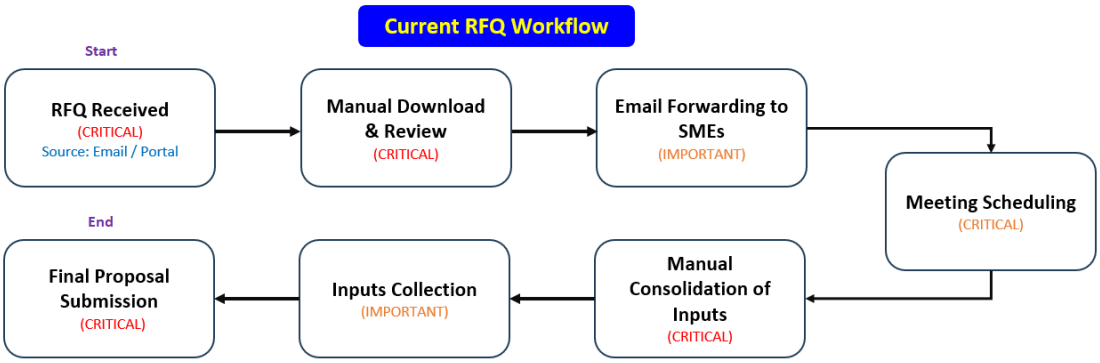
Why it matters:

- RFQs often represent high-value opportunities (€10M+). Inefficiencies reduce profitability, increase risk, and impact customer satisfaction.
- **Complexity of RFQ Process (Supporting Example)** While the main issue is manual coordination, complexity amplifies the challenge. For instance, a recent €20M proposal included a 50-page main document, 20 annexures, and multiple commercial breakdowns. Such cases highlight why manual processes are error-prone and time-consuming, reinforcing the need for AI-driven efficiency.

3. Current Process & Knowledge Pathways

How the task is performed today:

- RFQs received via email or OEM portals and manually reviewed.
- Inputs collected via email in multiple formats (Word, Excel, PDF, CAD, JSON).
- Multiple meetings for clarifications → miscommunication and delays.
- Manual consolidation of technical and commercial inputs into proposals.



Rules and heuristics:

- Prioritization based on perceived client importance.
- Cost estimation relies on historical experience and spreadsheets.

Operational Risks in Current Workflow			
Risk	**Cause**	**Impact**	**Severity**
Missed deadlines	Manual tracking	Lost revenue	High
Incorrect estimation	Wrong interpretation of data	Profitability hit	High
Miscommunication	Too many meetings	Rework	Medium
Incomplete inputs	Delayed SME responses	Delay in proposal	High
Poor proposal quality	Wrong assumptions	Customer dissatisfaction	High

Ambiguities and manual dependencies:

- Subjective prioritization → missed opportunities.
- Incomplete inputs → rework and delays.
- Accountability gaps due to multiple "consulted" roles.

Role & Responsibility Mapping (RASIC)						
Activity	**Engagement Manager**	**SME**	**Finance**	**Procurement**	**Sales**	**Delivery**
RFQ Review	A	C	I	I	R	C
Technical Inputs	I	R	C	C	I	S
Cost Estimation	A	C	R	C	C	C
Proposal Consolidation	R	C	I	I	C	S
Commercial Strategy	C	C	C	C	R	C

4. Data & Information Landscape (High-Level)

Types of data generated:

- RFQ documents (Word/PDF) – unstructured.
- CAD files – large size.
- JSON/CSV – inconsistent formats.
- Excel – manual updates.
- Email threads – hard to track.

Data & Information Landscape			
Data Type	**Source**	**Importance**	**Challenge**
Word/PDF	RFQ docs	High	Unstructured
CAD	OEM portal	High	Large size
JSON/CSV	Portal/system	Medium	Format inconsistency
Excel	Cost estimation	High	Manual updates
Email threads	Communication	High	Hard to track

Challenges:

- Unstructured data makes requirement extraction difficult.
- Inconsistent formats across OEMs.
- No centralized repository for RFQ-related data.
- Limited real-time visibility into progress.

5. Early AI Opportunity Hypothesis

How AI can help:

- **Document Intelligence**: Extract key requirements from RFQs, CAD files, JSON/CSV.
- **Workflow Automation**: Intelligent routing of RFQs to SMEs based on expertise.
- **Accountability Tracking**: Dashboards for ownership and deadlines.
- **Prioritization & Alerts**: Predictive prioritization based on client value and complexity.
- **Knowledge Reuse:** Suggest similar past RFQs and responses.

What success looks like:

- Reduce RFQ turnaround time by 30–40%.
- Improve proposal accuracy and completeness.
- Enhance accountability and transparency.
- Increase win rate for multi-million Euro RFQs.

Current vs Future Workflow Comparison

Aspect	Current Process	Future AI-Enabled Process
RFQ Intake	Manual download from email/portal	Automated intake & parsing
Data Handling	Unstructured, manual interpretation	AI-driven document intelligence
Task Assignment	Manual forwarding to SMEs	Intelligent routing based on expertise
Meetings	Multiple, time-consuming, miscommunication	Reduced, dashboard-based collaboration
Inputs Collection	Manual, multiple formats	AI-assisted, pre-filled templates
Consolidation	Manual, error-prone	Automated with human validation
Accountability	No clear tracking	Real-time dashboards & alerts
Effort	~500 hours for large RFQs	~300 hours with automation
Risk	High (missed deadlines, wrong estimation)	Lower (AI checks & structured workflow)

Note: The comparison represents an initial conceptual view of the future AI-enabled workflow. This will be refined and expanded in subsequent phases as feasibility, ecosystem mapping, and governance considerations are addressed.

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6. Academic Declarations

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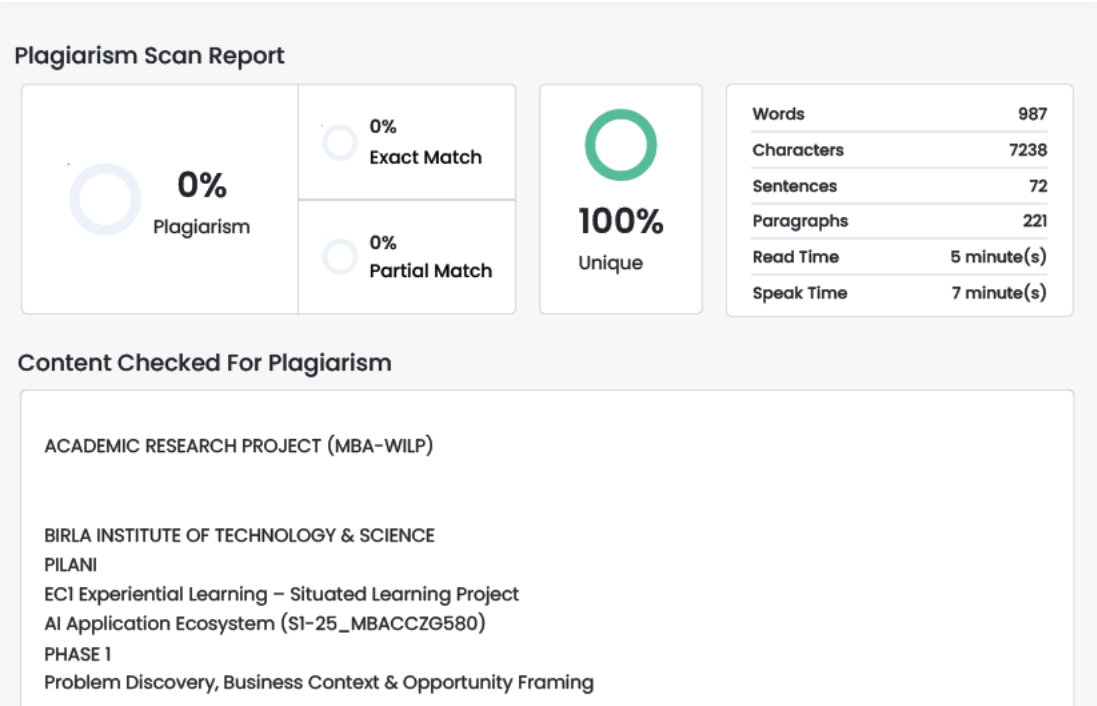

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